ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Mechatronics paper 1 · practice d.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, sensible design choices and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 B ✓ 1.1.3 A ✓ 1.1.4 C ✓ 1.1.5 B ✓ 1.1.6 D ✓ 1.1.7 A ✓ 1.1.8 C ✓ 1.1.9 B ✓ 1.1.10 A ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 False ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 False ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 G ✓ 1.3.3 I ✓ 1.3.4 E ✓ 1.3.5 A ✓ 1.3.6 J ✓ 1.3.7 F ✓ 1.3.8 H ✓ 1.3.9 B ✓ 1.3.10 D ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Infeed conveyor → filler → capper → labeller → inspection camera with reject unit → packer, in the correct order ✓✓; conveyors linking the machines ✓; at least TWO sensors shown, e.g. bottle-present sensors, fill-level check, cap-present sensor ✓; neat and labelled ✓ (5)

2.2 $\text{Bottleneck: the capper (65 bottles per minute)} \checkmark; \text{output} = 65 \times 60 \checkmark = 3\,900 \text{ bottles per hour} \checkmark; 3\,900 > 3\,600, \text{so the line meets the target} \checkmark$ (4)

2.3 Steps in order with confirming signals ✓✓✓✓✓✓, e.g.: (1) stopper extends – stopper-out sensor; (2) bottle arrives at the stopper – bottle-present photoelectric sensor; (3) nozzle lowers – nozzle-down sensor; (4) valve opens until the volume is reached – flow meter count / timer done; (5) valve closes and nozzle rises – nozzle-up sensor; (6) stopper retracts and the bottle moves on – stopper-in sensor / bottle-gone sensor. (6)

2.4 $t = V \div Q \checkmark = 0{,}5 \div 0{,}25 \checkmark = 2 \text{ s} \checkmark$ (3)

[18]

QUESTION 3

3.1 $v = \text{bottles per second} \times \text{pitch} \checkmark = (65 \div 60) \times 0{,}1 \checkmark = 0{,}11 \text{ m/s } (0{,}1083 \text{ m/s}) \checkmark$ (3)

3.2 $N = 60v \div (\pi D) \checkmark = 60 \times 0{,}1083 \div (\pi \times 0{,}1) \checkmark = 20{,}69 \text{ rpm} \checkmark$ (3)

3.3 $\text{Ratio} = 1\,400 \div 20{,}69 \checkmark = 67{,}67 : 1 \checkmark$ (2)

3.4 $T = F \times r = 300 \times 0{,}05$ ✓ $= 15$ N·m ✓; $P = F \times v = 300 \times 0{,}1083$ ✓ $= 32{,}50$ W ✓ (4)

3.5 $P_{in} = P_{out} \div (\eta_{gearbox} \times \eta_{motor})$ ✓ $= 32{,}5 \div (0{,}8 \times 0{,}85)$ ✓ $= 47{,}79$ W ✓ (3)

3.6 Any THREE ✓✓✓: the conveyor speed can be matched to the speed of the other machines; soft start and stop, so bottles do not fall over; the speed can be changed for different products from the HMI / recipe; saves energy; protects the motor (overload, phase loss). (3)

[18]

QUESTION 4

4.1 Any FOUR sensible inputs with addresses ✓✓✓✓, e.g. bottle at capper, cap present, cap torque OK, bottle at reject, camera fault signal, reject-bin full, emergency stop OK, guard closed. Any THREE outputs ✓✓✓, e.g. capping head motor, cap-feed motor, reject air valve, conveyor motor / VSD run, alarm beacon. (7)

4.2 When the camera finds a faulty bottle, the PLC stores a "fault" bit for that bottle ✓ in a shift register (or FIFO) ✓; a sensor or encoder pulse shifts the bits along by one place every time a bottle moves one pitch ✓; when the fault bit reaches the fifth position, the bottle at the reject unit is the faulty one and the air valve fires ✓. (4)

4.3 Rung 1: X6 AND M1 start timer T1 with a preset of 0,2 s ✓✓; rung 2: X6 AND M1 AND NC T1 switch on Y5, so Y5 goes off when T1 is done ✓✓, e.g. (4)

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|--[ X6 ]--[ M1 ]-----------------( T1  TON 0,2 s )--|
|--[ X6 ]--[ M1 ]--[/ T1 ]--------( Y5 )-------------|
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4.4 Any THREE ✓✓✓: emergency stop pressed; guard door open; conveyor jam / bottle fallen; low cap or label supply; reject bin full; motor overload / VSD fault; low air pressure; too many rejects in a row; water tank level low. (3)

[18]

QUESTION 5

5.1 Four items, each with a sensible task ✓ and frequency ✓, e.g.: conveyor belt – check tension, tracking and wear – weekly; gearbox – check the oil level / change the oil – monthly / every 2 000 h; filler nozzles and valves – clean and sanitise (CIP) – daily; pneumatic system – drain the water trap, check for leaks – daily / weekly; sensors and camera – clean lenses, test function – daily; capper chucks – check grip and torque – weekly; motor bearings – grease, vibration check – monthly. (8)

5.2 Method ✓ and what it detects ✓ each, any THREE: vibration analysis – worn or unbalanced bearings, misalignment; thermal imaging – overheating motors, bearings or loose electrical connections; oil analysis – metal particles from wear, contaminated oil; motor current monitoring – overloads, jams, worn gears; ultrasonic detection – air leaks, failing bearings. (6)

5.3 The line handles drinking water, so dirt, bacteria or ordinary oil could contaminate the product and make people ill ✓; food-grade lubricants are safe if they touch the product and the factory must meet food-safety standards ✓. (2)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE